

Pre-Practicum 3: NI ELVIS

(EBN 111 : Electricity and Electronics)

Introduction to: Operational Amplifier Circuits

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Agenda



- ☐ Objectives
- ☐ Description of Operational Amplifiers
- ☐ Physical Construction on NI Elvis prototyping board
- ☐ Physical connections of the Operational Amplifier circuit
- ☐ Practical Preparation

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Objectives



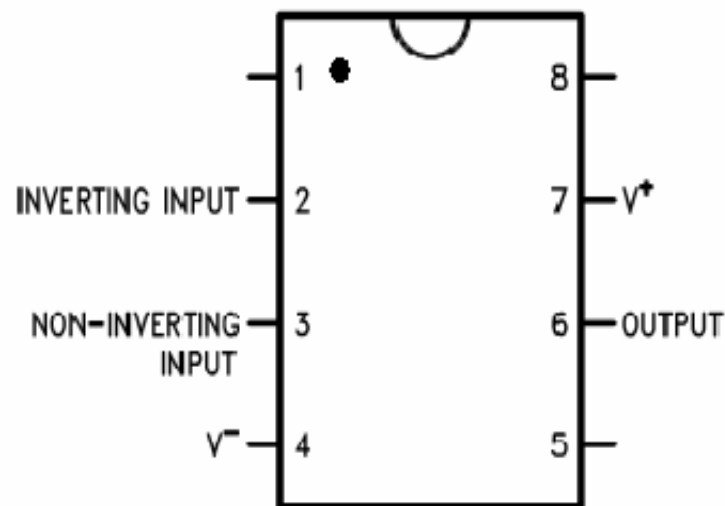
- ❑ The objective of this practical is to build two operational amplifier circuits, and
- ❑ To record their transfer characteristics
 - **i.e.** Input Voltage versus Output Voltage
 - **NB:** Read about the basic concepts of Inverting and Non-Inverting amplifiers in your textbook
- ❑ The following components will be provided during the practical
 - Two resistors, 1 k Ω and 2.2 k Ω
 - LM 741 Operational Amplifier
 - Connecting wires
- ❑ NI Elvis workstation and software will be used
- ❑ The same guidelines of practical 2 for launching the software and using the workstation should be followed

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Description of Op-Amps

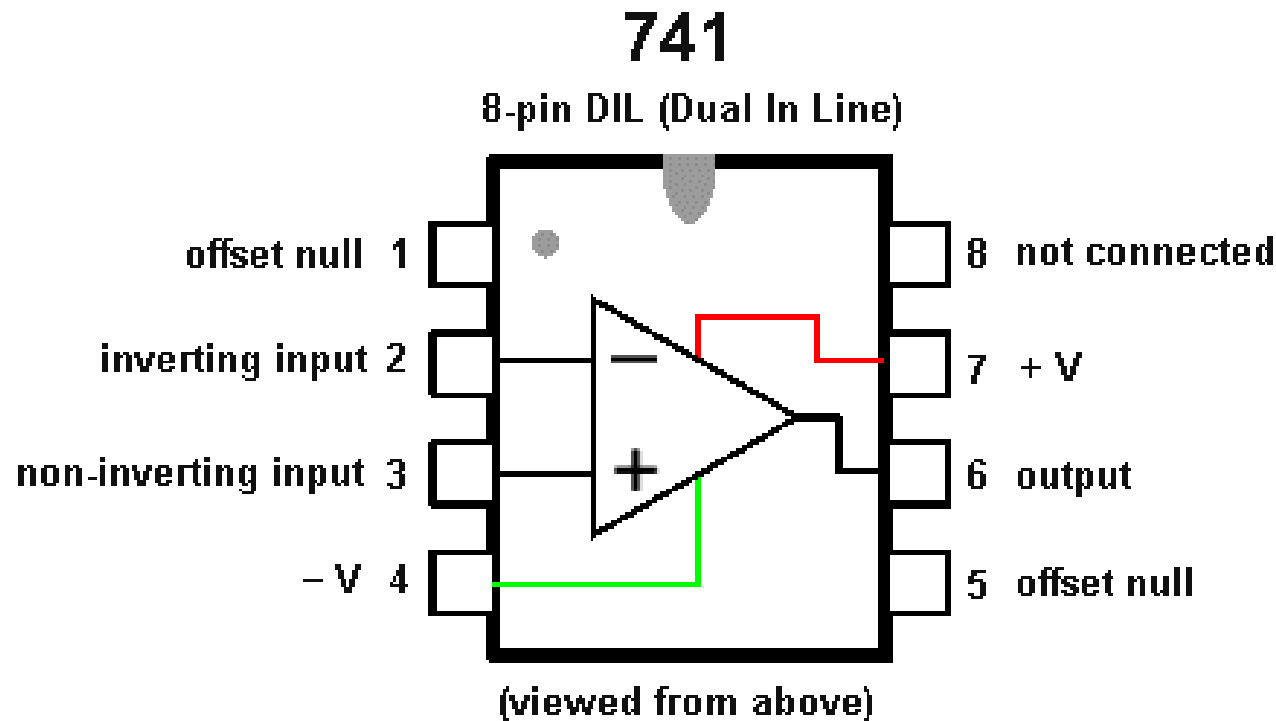


- LM741 Operational Amplifier will be used for this practical, it will be in 8 pin Dual In-Line Package
- The physical package layout of the Op-Amp is shown here
- The pins correspond to the various inputs and outputs of the Op-Amp
- Pin 1 is identified with the dot next to it and all other pins are counted counter-clockwise from pin 1



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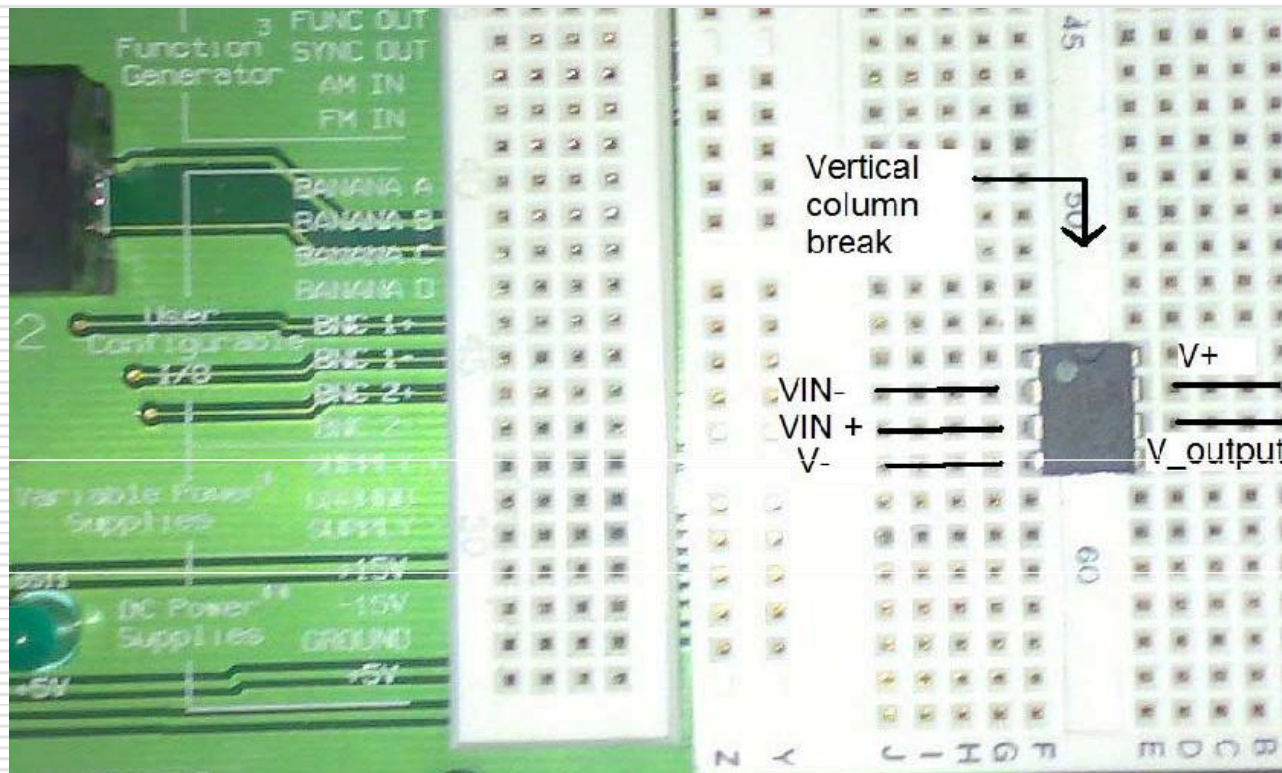


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Constructions on NI Elvis



- Place the Op-Amp on the prototyping board so that pin 4 and 5 are separated by the vertical column break on the board



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Physical Connections



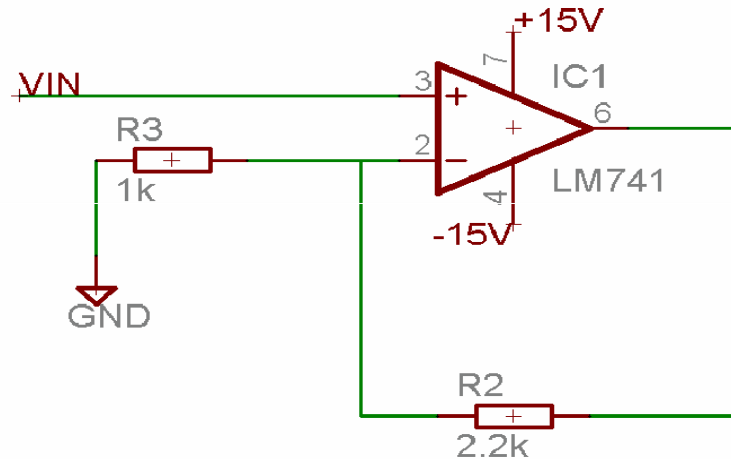
- ❑ Connect V+ (pin 7) to the +15 V power supply on the NI workstation
- ❑ Connect V- (pin 4) to the -15 V power supply on the NI workstation
- ❑ Connect the input signal to variable power supply (VPS) on the NI workstation
- ❑ All grounds must be connected to the ground of the NI workstation
- ❑ Connect resistors according to specific question (1 or 2)
- ❑ Two circuits will be constructed:
 - Inverting Amplifier
 - Non-Inverting Amplifier

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Practical Preparation



- Prepare for the practical by using the following questions and figure below:
 - Is the output inverted from the input?
 - What is the theoretical gain of the amplifier?
 - What is the relationship between input and output voltage?
 - Amplification, knee-voltage and saturation regions

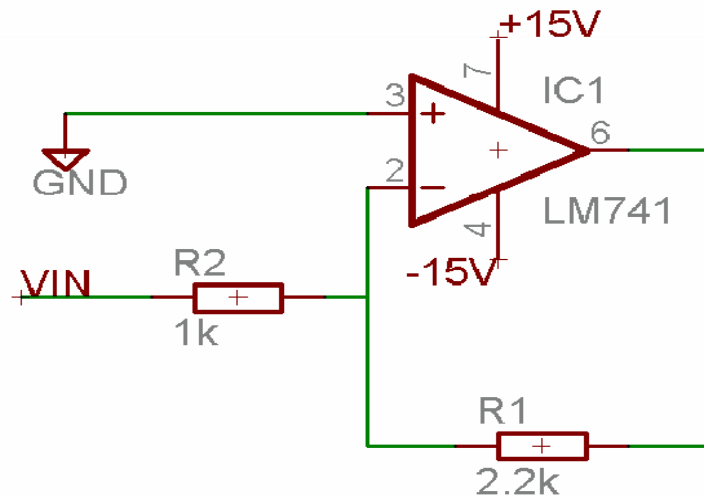


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Practical Preparation



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Conclusion



□ Contact details for Questions or Feedback

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□ During practical

- There will be Demis to assist you during the whole duration of the practical
- Ask questions and remember you can also refer to the practical guide or materials available on the module website